Product Requirements Document: VoteHub

Working name: **VoteHub** · ISM 6427 · Version 1.0 · October 27, 2026 · Owner: Cecilia Carvalho

1. Overview

VoteHub is a single web app where a voter types in their address and gets everything they need to vote: where to vote, what's on their ballot, key dates and reminders, their districts, and campus-specific help for FAU students. It's built on the Google Civic Information API v2, has **no database**, and deploys to **Netlify**.

> **Design constraints that shape everything below:**
>
> - **No database.** All user data (address, saved ballot picks, reminder choices) lives in the browser's `localStorage`. Nothing is stored on a server.
> - **Netlify.** Static frontend plus **Netlify Functions** as a thin proxy, so the Google API key never reaches the browser.
> - **Email/SMS reminders are out of scope.** They require storing users' contact info and a scheduler, which means a database. Reminders are delivered as calendar files (.ics) and "Add to Google Calendar" links instead.
> - **No representatives data.** Google retired that endpoint in April 2025. The app shows districts only; officials are a stretch goal via a third-party API.

2. Goals & non-goals

Goals	Non-goals (v1)
• A voter can find their polling place, early-vote sites, and drop boxes in under 30 seconds • A voter can build and print a personalized sample ballot • A voter can add every key election date to their calendar in one click • A voter understands which districts they live in • FAU students get a tailored on-ramp	• User accounts or login • Any server-side storage • Email or SMS reminders • Voter registration submission (we link to the state) • Candidate endorsements or recommendations (the app stays neutral)

3. Target users

- **General voter:** wants to know where and when to vote and what's on the ballot.
- **First-time / student voter (FAU):** unsure of registration, absentee rules, and polling location near campus.
- **Prepared voter:** wants to research races and bring a checklist to the polls.

4. Features & requirements

F1. "Where do I vote?" finder

- Address input with Google Places Autocomplete; the address is saved to `localStorage` for return visits.
- Calls `voterInfoQuery`. Shows three tabs: **Election Day polling place**, **Early voting sites**, **Drop-off locations**, each with address, hours, dates, and notes.
- Google Map (Maps JavaScript API) with color-coded pins; each pin has a "Get directions" link that opens Google Maps.
- Empty state: if no polling data comes back, show the county Supervisor of Elections link from `state[].local_jurisdiction` and explain that data appears closer to the election.

F2. Personalized sample ballot

- Renders every `contest` from `voterInfoQuery`, grouped by level: Federal → State → County → Local → Referenda.
- Each race shows office, district, and candidates (name, party, website, social links). Each referendum shows title, subtitle, and pro/con text.
- Voter can mark a pick per race (or "undecided"). Picks are saved to `localStorage`, keyed by `electionId`.
- **Print / Save as PDF** button produces a clean one-page checklist via a print stylesheet.
- Neutral display: candidates in the order the API returns them, no ratings or endorsements.

F3. Election countdown & reminders

- Calls `electionQuery` and filters to elections whose `ocdDivisionId` matches one of the user's districts (from F4).
- Countdown card on the home screen: days until the next election, plus early-voting start/end dates when the API returns them.
- "Add to calendar" for each date: a downloadable **.ics file** (generated client-side) and an **Add to Google Calendar** link. The .ics includes a 1-day-before alarm.
- Optional browser notification while the tab is open (Notification API). No background push in v1.

F4. "Know your districts" explainer

- Calls `divisionsByAddress` and converts each OCD-ID into plain language, e.g. `ocd-division/country:us/state:fl/cd:23` → "U.S. House District 23."
- A small lookup table in the frontend maps OCD type codes (`cd`, `sldu`, `sldl`, `county`, `place`, `school_district`) to friendly labels.
- Each district gets a short "what this body does" tooltip.
- **Stretch:** show officials by passing OCD-IDs to a third-party API (Cicero or similar) through another Netlify Function.

F5. FAU campus voter hub

- Dedicated `/fau` page with quick-pick buttons for FAU campuses and residence halls (Boca Raton, Jupiter, Davie, etc.) that prefill an address and run F1.
- Shows the registration, absentee/vote-by-mail, and election-rules links returned in `state[].electionAdministrationBody`.
- A static "Student voting FAQ" (registering at a school vs. home address, needed ID, vote-by-mail), with each answer linking to the official Florida source rather than stating rules from memory.
- Shareable link and QR code for posters and class announcements.

5. Architecture

```
Browser (static HTML/CSS/JS or React via Vite)
  │  localStorage: address, ballot picks, settings
  │
  ├── /.netlify/functions/elections   → GET civicinfo/v2/elections
  ├── /.netlify/functions/voterinfo   → GET civicinfo/v2/voterinfo?address&electionId
  ├── /.netlify/functions/divisions   → GET civicinfo/v2/divisionsByAddress?address
  │        (each function adds GOOGLE_CIVIC_API_KEY from Netlify env vars)
  │
  └── Google Maps JS + Places Autocomplete (browser key, HTTP-referrer restricted)
```

Component	Choice	Why
Frontend	React + Vite (or plain JS)	Netlify builds it natively; simple routing for 5 pages
Backend	3 Netlify Functions (Node)	Hides the Civic API key; no server to run
Storage	`localStorage` only	No database requirement
Caching	`Cache-Control` headers on function responses (e.g., elections 6h, voterinfo 1h)	Cuts API calls and quota use
Secrets	Netlify environment variables	Key never committed to Git
Calendar	Client-side .ics generation	Reminders without storing anything

6. Pages & navigation

Route	Content
`/`	Address entry, countdown card, shortcuts to each feature
`/where`	F1: map + polling, early, and drop-off tabs
`/ballot`	F2: sample ballot with picks + print
`/dates`	F3: election dates + calendar exports
`/districts`	F4: district explainer
`/fau`	F5: campus hub

7. Non-functional requirements

- **Privacy:** the address is sent only to our functions and Google; it's never logged. A "Clear my data" button wipes `localStorage`.
- **Accessibility:** WCAG 2.1 AA, keyboard navigable, screen-reader labels on the map list view (a text list must always accompany the map).
- **Mobile-first:** most use happens on phones near Election Day.
- **Performance:** first results in under 3 seconds on 4G.
- **Neutrality:** no partisan content, no endorsements, candidates shown in API order.
- **Error handling:** clear messages for an invalid address, no active election, API quota exceeded, and network errors.

8. Risks & mitigations

Risk	Mitigation
`voterinfo` is empty outside election windows	Graceful empty states; link to county elections office; demo with `electionId=2000` (Google's test election)
Incomplete local data for some counties	Always show the official state/county links as a fallback
API key exposure	Key only in Netlify Functions; Maps key restricted by referrer
localStorage cleared or unavailable	App works without it; picks are a convenience, not required
No representatives endpoint	Districts only in v1; officials as a third-party stretch goal

9. Success metrics

- Address-to-polling-place result shown in under 30 seconds for 90% of test users
- Sample ballot printed or saved by at least 25% of users who open it
- Calendar file downloads per unique session
- Zero API keys exposed in client bundles (checked before each deploy)

10. Milestones (fast build)

Phase	Scope
1. Setup	Netlify site, env var for key, 3 proxy functions, address input
2. Core	F1 polling finder with map; F2 sample ballot with picks + print
3. Extend	F3 dates + .ics; F4 districts explainer
4. Campus + polish	F5 FAU hub, accessibility pass, empty/error states, deploy

11. References

- Google Civic Information API v2: https://developers.google.com/civic-information/docs/v2
- voterInfoQuery: https://developers.google.com/civic-information/docs/v2/elections/voterInfoQuery
- divisionsByAddress: https://developers.google.com/civic-information/docs/v2/divisions/divisionsByAddress
- Representatives API turndown notice: https://groups.google.com/g/google-civicinfo-api/c/9fwFn-dhktA/m/ftUoL83mAwAJ
- Netlify Functions: https://docs.netlify.com/functions/overview/